Assignment 1

BSSE23058

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Section A

Question 1: Decision Tree Construction (AND Function)

Tasks:

1. Create the full dataset (all 8 combinations).

x1	x2	x3	y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

2. Using Information Gain, determine: The best feature for the root node.

$H(D)$: Dataset has 7 instances of '0' & 1 instance of '1'

$$H(D) = -\left(\frac{7}{8}\log_2\frac{7}{8} + \frac{1}{8}\log_2\frac{1}{8}\right) \approx 0.5435$$

Info Gain for x_1:

When $x_1 = 0$ (4 instances): All y values are 0.

Entropy $H(D_{x_1=0}) = 0$. $y\{0,0,0,0\}$

When $x_1 = 1$ (4 instances): y values are $\{0,0,0,1\}$

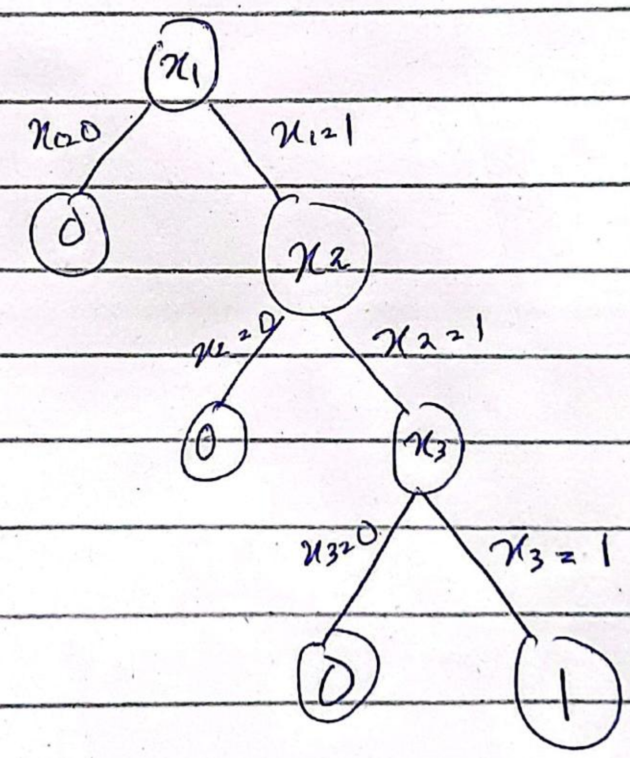
$$H(D_{x_1=1}) = -\left(\frac{3}{4}\log_2\frac{3}{4} + \frac{1}{4}\log_2\frac{1}{4}\right) \approx 0.8113$$

$$IG(D, x_1) = 0.5435 - \left(\frac{4}{8}(0) + \frac{4}{8}(0.8113)\right)$$

$$= 0.5435 - 0.4056$$

$$= 0.1379$$

3. Draw the final tree clearly.



Question 2: Gradient Descent on a Quadratic Loss

Tasks:

1. Compute the gradient of the loss function.
2. Perform two iterations of gradient descent:
   - Compute $w1$
   - Compute $w2$
3. Compute the loss at each step: $L(w0)$,$L(w1)$,$L(w2)$

$L(w) = (w-3)^2 + 2$

$\frac{dL}{dw} = 2(w-3) \times 1 + 0$

$= 2(w-3)$

$\alpha = 0.1 \quad w0 = 0 \qquad w_{new} = w_{old} - \alpha \nabla L(w_0)$

$\nabla L(w_0) = 2(0-3) = -6$

$w_1 = 0 - (0.1)(-6) = 0.6$

$\nabla L(w_1) = 2(0.6-3) = 2(-2.4) = -4.8$

$w_2 = 0.6 - (0.1)(-4.8) = 1.08$

$L(w_0) = (0-3)^2 + 2 = 9+2 = 11$

$L(w_1) = (0.6-3)^2 + 2 = (-2.4)^2 + 2$

$= 7.76$

$L(w_2) = (1.08-3)^2 + 2 = (-1.92)^2 + 2$

$= 5.6864$

Question 3: K-Means Clustering (Unsupervised Learning)

Tasks:

1. Assign each data point to the nearest centroid using Euclidean distance.

Points $A(1,1)$, $B(2,1)$, $C(4,3)$, $D(5,4)$

Centroids $\mu_1(1,1)$

$\mu_2(5,4)$

$A \rightarrow \mu_1 = \sqrt{(1-1)^2 + (1-1)^2} = 0$

$A \rightarrow \mu_2 = \sqrt{(5-1)^2 + (4-1)^2} = 5$ Cluster 1

$B \rightarrow \mu_1 = \sqrt{(1)^2 + (0)^2} = 1$

$B \rightarrow \mu_2 = \sqrt{3^2 + 3^2} = 4.24$ Cluster 1

$C \rightarrow \mu_1 = \sqrt{3^2 + 2^2} = 3.61$

$C \rightarrow \mu_2 = \sqrt{1^2 + 1^2} = 1.41$ Cluster 2

$D \rightarrow \mu_1 = 5$

$D \rightarrow \mu_2 = 0$ Cluster 2

2. Compute the new centroids after reassignment.

New centroids

$$\mu_{1,new} = \left(\frac{1+2}{2}, \frac{1+1}{2}\right) = (1.5, 1)$$

$$\mu_{2,new} = \left(\frac{4+5}{2}, \frac{3+4}{2}\right) = (4.5, 3.5)$$

3. Perform one more iteration (assignment + update).

A(1,1) is closer to μ_1 than μ_2 Cluster 1
B(2,1) is closer to μ_1 than μ_2 Cluster 1
C(4,3) is closer to (μ_2) than μ_1 Cluster 2.
D(5,4) is closer to μ_2 than μ_1 Cluster 2.

4. Report the final cluster assignments.

Centroids remain (1.5, 1) & (4.5, 3.5)
as the membership didn't change.

Cluster 1: {(1, 1), (2, 1)}
Cluster 2: {(4, 3), (5, 4)}.